Carr and Wu (2003): What Type of Process Underlies Options? A Simple Robust Test

Introduction

A fundamental question in asset pricing is whether stock prices are best described by a purely continuous process (PC), a pure jump process (PJ), or a combination of both (CJ). The direct approach — examining time-series data — is unreliable unless the sampling frequency is extremely high, and high-frequency data introduce microstructure noise. Carr and Wu (2003) sidestep this problem entirely by looking at option prices instead. Their key insight is that, although discretely sampled paths from the three process types look nearly identical, they have dramatically different implications for short-maturity option prices.

The Core Method: Term Decay Plots

Under a risk-neutral measure, the asset price dynamics are modelled as:

$$\frac{dS_t}{S_{t-}} = (r-q)\,dt + \sigma_t\,dW_t + \int_{\mathbb{R}^0} (e^x - 1)\big[\mu(dx, dt) - \nu_t(x)\,dx\,dt\big] \tag{1}$$

where W_t is a standard Brownian motion, $\mu(dx, dt)$ counts jumps of log-size x, and $\nu_t(x)$ is the compensator (local jump density). The first stochastic term is the purely continuous martingale component; the integral is the purely discontinuous (jump) martingale component.

The paper's main theoretical result decomposes the time value of a European option into two parts — one driven by the continuous component and one by the jump component:

$$TV_0(K,T) = e^{-rT} \int_0^T \Big[\tfrac{1}{2} q(K,t) K^2\, \mathbb{E}_0[\sigma_t^2 \,|\, F_{t-} = K] + \mathbb{E}_0[F_{t-}\,\nu_t^0(k)]\Big]\,dt \tag{2}$$

where $q(K,t)$ is the risk-neutral density of the forward price, $k = \ln(K/F_{t-})$ is log-moneyness, and $\nu_t^0(k)$ is the double tail of the jump compensator. As $T \downarrow 0$, this decomposition reveals that the speed at which option prices converge to zero depends entirely on which components are present, as summarised in Table 1.

Table 1: Asymptotic Decay Rates of Short-Maturity Option Prices

Process Type	OTM Options	ATM Options
Purely Continuous (PC)	$O(e^{-c/T}),\ c > 0$	$O(\sqrt{T})$
Pure Jump (PJ)	$O(T)$	$O(T^p),\ p \in (0, 1]$
Combined (CJ)	$O(T)$	$O(T^p),\ p \in (0, \frac{1}{2}]$

The intuition is straightforward. Under a purely continuous process, an out-of-the-money option can only move in the money through gradual diffusion, which is exponentially unlikely over a very short horizon. When jumps are present, however, the stock can leap across the strike in a single instant, so OTM option prices decay only linearly in T. For ATM options, the $O(\sqrt{T})$ rate signals an infinite-variation component (either diffusion or an infinite-variation jump process), while $O(T)$ signals a pure finite-variation jump process.

To apply the test empirically, the authors plot $\ln(P/T)$ against $\ln(T)$, calling this a *term decay plot*. A flat slope near zero for OTM options signals a jump component; a strongly positive, concave curve signals a purely continuous process. For ATM options, a straight line with slope near -0.5 signals an infinite-variation component; a flat line signals finite-variation jumps only.

Empirical Findings: S&P 500 Index Options

Carr and Wu apply the method to daily S&P 500 index option data from April 1999 to May 2000 (273 business days). They fit a second-order polynomial,

$$\ln(P/T) = a(\ln T)^2 + b(\ln T) + c, \tag{3}$$

to the term decay plot at each date and moneyness level, extracting slope and curvature estimates. The results clearly support a combined continuous-jump (CJ) process. OTM term decay plots are mostly flat at short maturities (slopes near zero), confirming the presence of a jump component. ATM term decay plots are consistently negatively sloped and approximately linear with slopes averaging -0.413, close to the -0.5 theoretical value for a continuous component, confirming that an infinite-variation component is always present. Importantly, the jump component varies substantially over time — on some days it is almost undetectable — while the continuous component is constantly and stably felt throughout the sample.

Strengths, Limitations, and Relevance to QQQ Options

The main strength of the method is its robustness: it does not require a parametric model and is not confined to a Markovian or single-factor framework. It can detect the type of process directly from observable option prices, without needing high-frequency return data. The simulation evidence confirms that the asymptotic behavior is visible for options maturing within 20 days, which are readily available in liquid markets.

The main limitation is that the test cannot always distinguish between a continuous martingale and an infinite-variation pure jump process, since both generate a similar $O(\sqrt{T})$ ATM decay rate. The method also operates under the assumption that the compensator σ_t

and ν_t are bounded near $t = 0$; deterministically explosive volatility could mimic jump-like decay even in a purely continuous model.

For our project on QQQ options, the Carr and Wu framework provides direct empirical motivation for including a jump component in the pricing model. The NASDAQ-100's technology-heavy composition makes it prone to sudden, large price moves, and if QQQ option prices exhibit the flat short-maturity OTM decay pattern documented here, this constitutes market-based evidence that jumps are priced into the index. The finding that the continuous component is always present further supports the use of combined jump-diffusion models — such as Merton (1976) or Kou (2002) — rather than pure jump specifications. The time-variation in the jump component also suggests that models with stochastic jump intensity may be necessary for capturing the full dynamics of the QQQ implied volatility surface across different market regimes.